封面

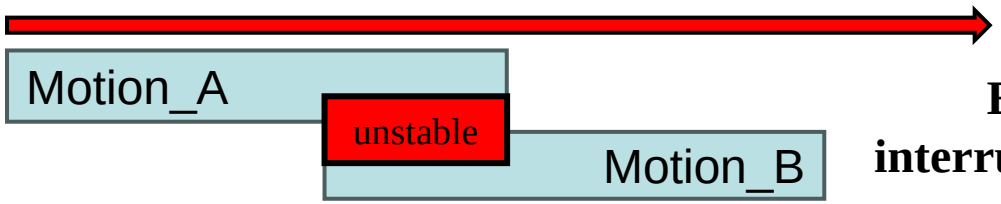
Stricker的升级

Defender的升级

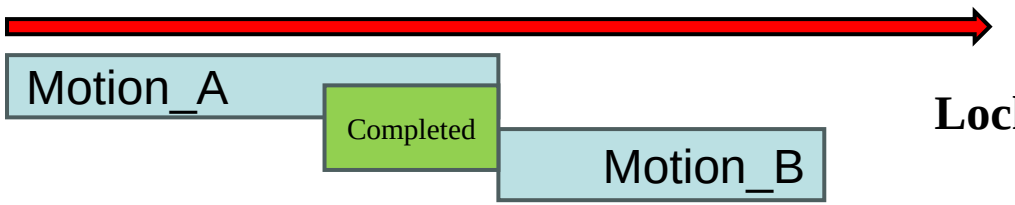
Goalkeeper的升级

motion的升级

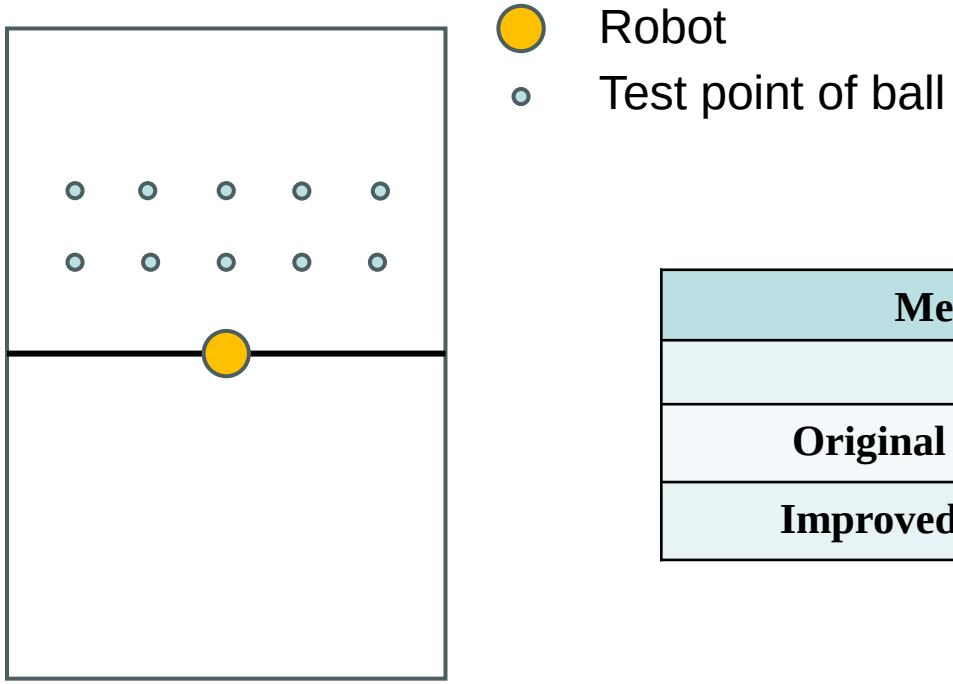
Motion Upgrade I: Reliability



Execution of ongoing motions was interrupted by fluctuations in external states.

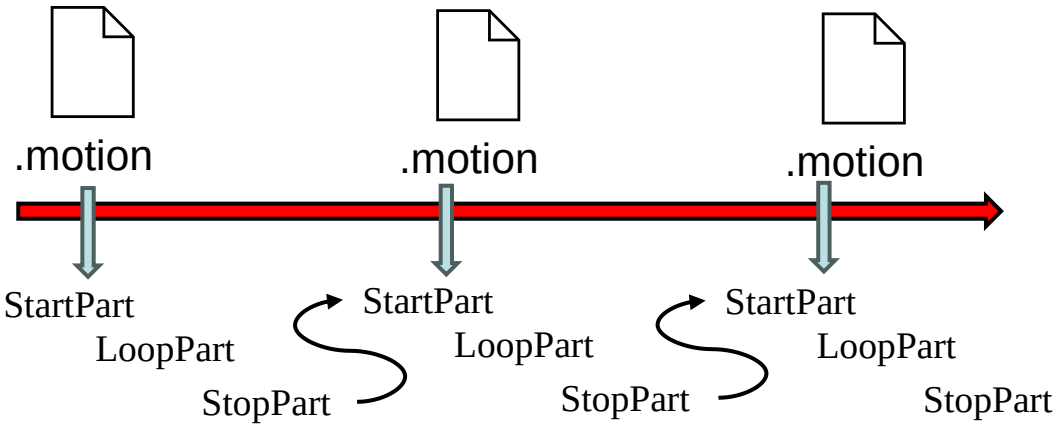


Lock current motion until completion to prevent premature switching.



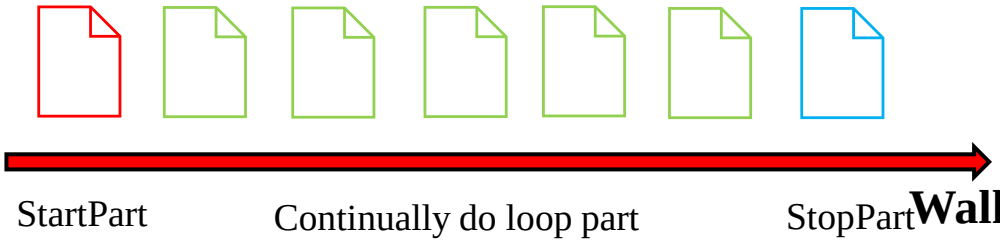
Metric : Failure Rate	
	Stuttering & Instability
Original	19/30
Improved	0/30

Motion Upgrade II: Speed & Smoothness



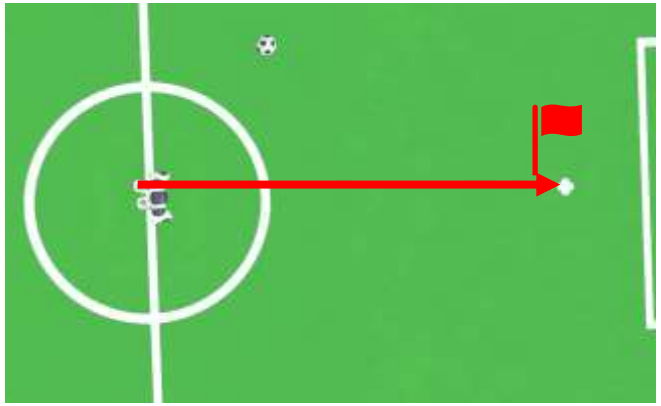
Redundant "Start" and "Stop" phases for every single file.

Frequent transitions lead to significant time overhead



Single StartPart and StopPart for the entire sequence.

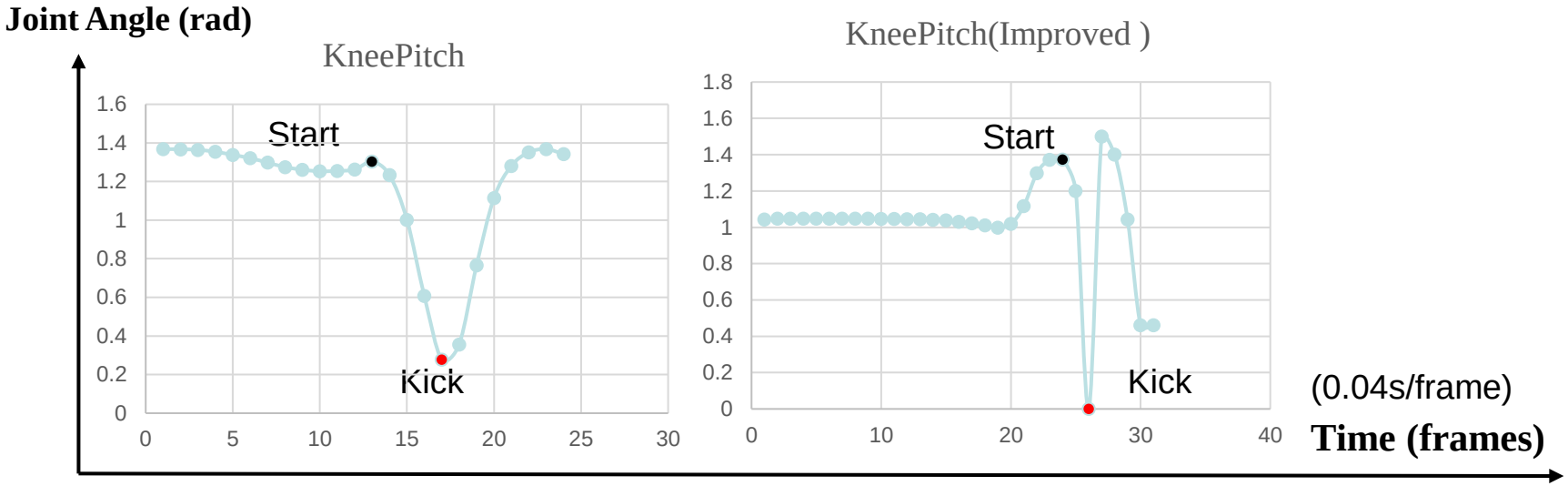
Walking Speed Improvement



Metric : walking time(Seconds)	
	Time to Cross Midfield
Original	94
Improved	43

-54%

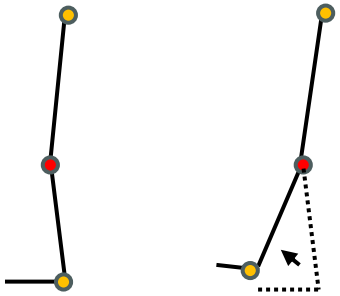
Motion Upgrade III:Kinematic Optimization



4 frames from 1.3 to 0.27

2 frames from 1.4 to 0

2x faster knee extension velocity.

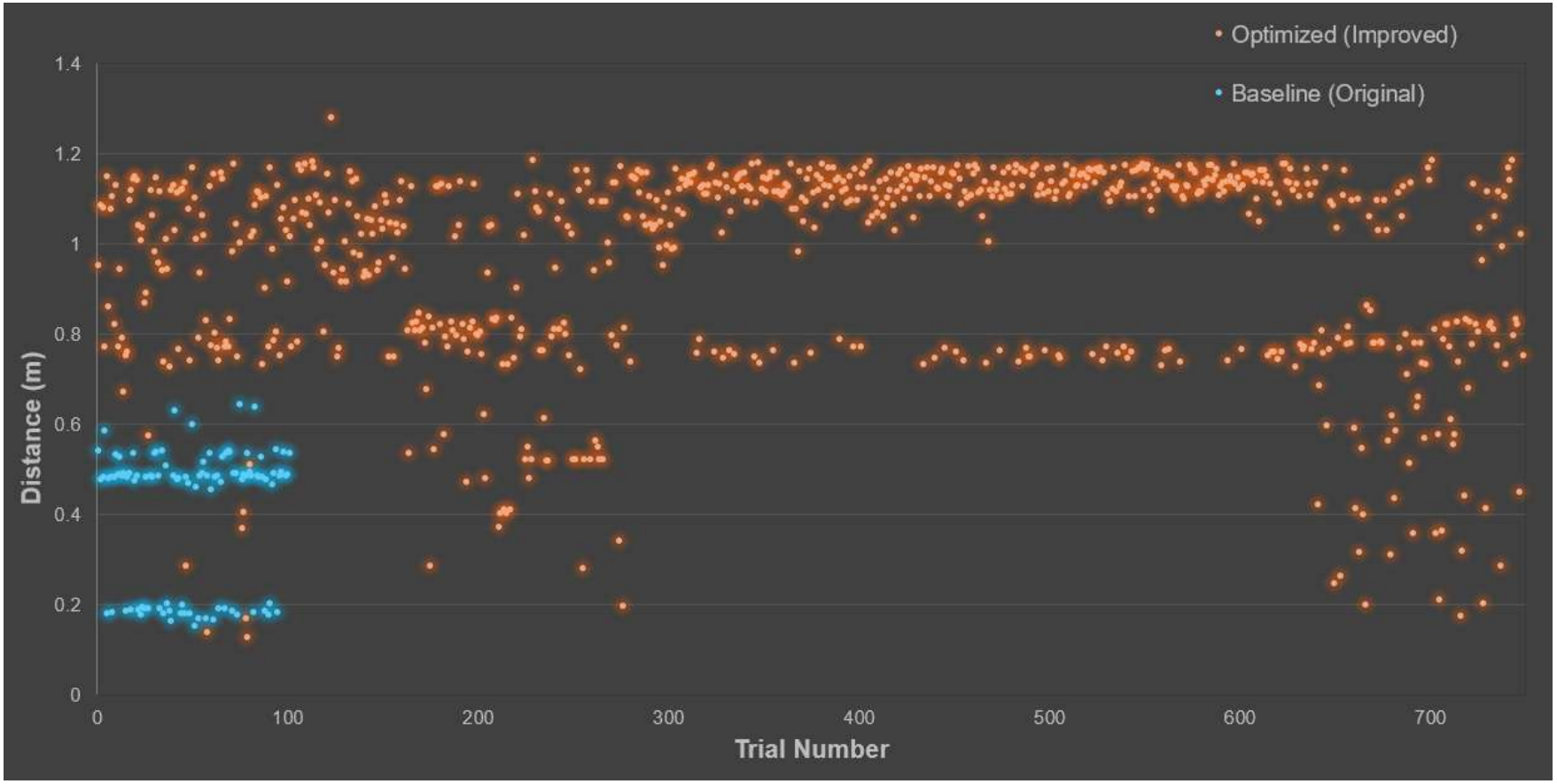


Quickly extend knee forward to kick the ball

Metric : Kicking distance(meters)			
	Training Mean(m)	Training Max(m)	Match Reliability
Original	0.3694	0.644	unstable
Improved	0.9228	1.2792	Consistently Stable

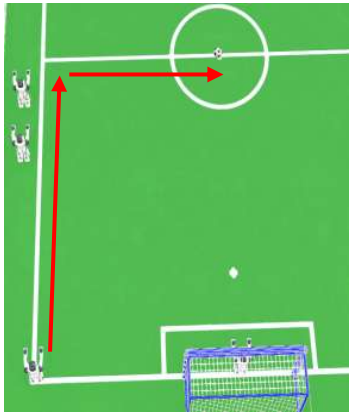
HipPitch and AnklePitch are also important!

Appendix: Kicking Distance Data

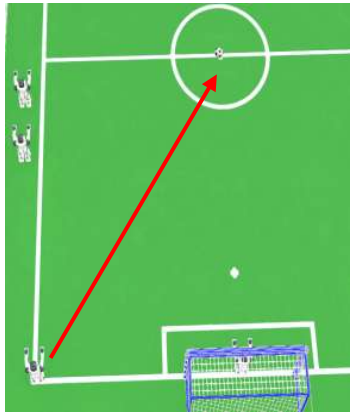


Performance Breakthrough: Large-scale Robustness Validation

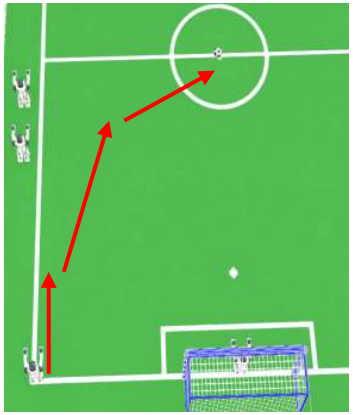
Striker Optimization: Path Planning & Efficiency



Baseline (L-Path)

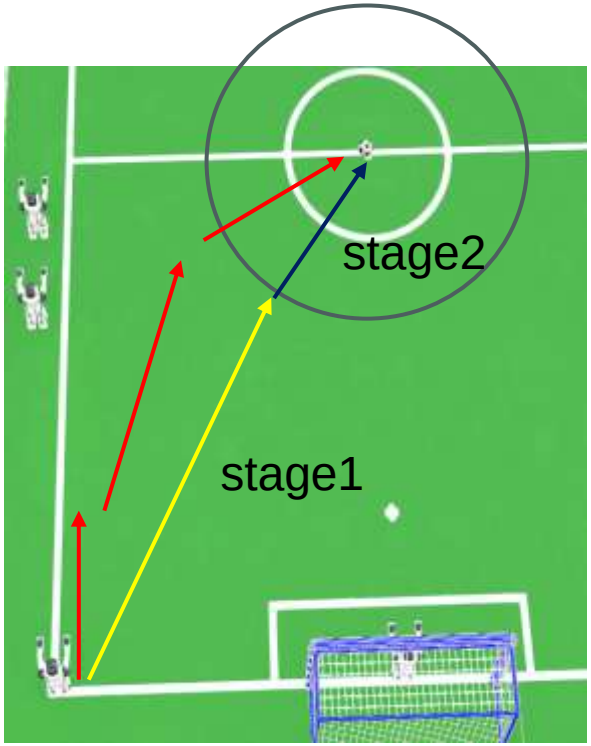


Linear Path (Diagonal)



Multi-stage Path (Optimized)

Metric : Travel Time to Ball (Seconds)		
Path	Static Target	Dynamic Target (Slight Movement)
Baseline (L-Path)	240	-
Linear Path (Diagonal)	108.6	137
Multi-stage Path (Optimized)	112	117.4



Stage1:Minimizing heading adjustments，Maintaining maximum approach velocity（Robustness against target displacement）
Stage2:Precision striking，Prioritizing angular correction over displacement

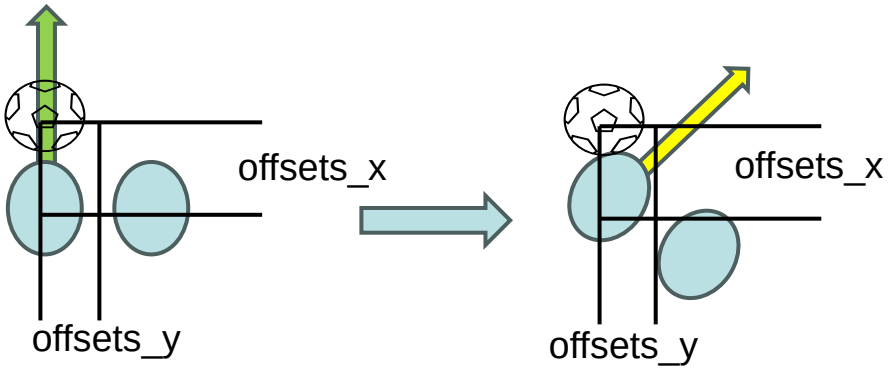
Striker Optimization: high-precision striking

Stage 1: Fixed Parameters
Static offsets leading to high error rates at angles.

Stage 2: Data-Driven Lookup
Supervised learning to record optimal (dx,dy) for specific angles.

Stage 3: Trigonometric Fitting (Optimized)
Generalized formulas for real-time, precise offset calculation.

```
dx = 0.27 * math.cos(deg + 0.3)
dy = 0.27 * math.sin(deg + 0.3)
```



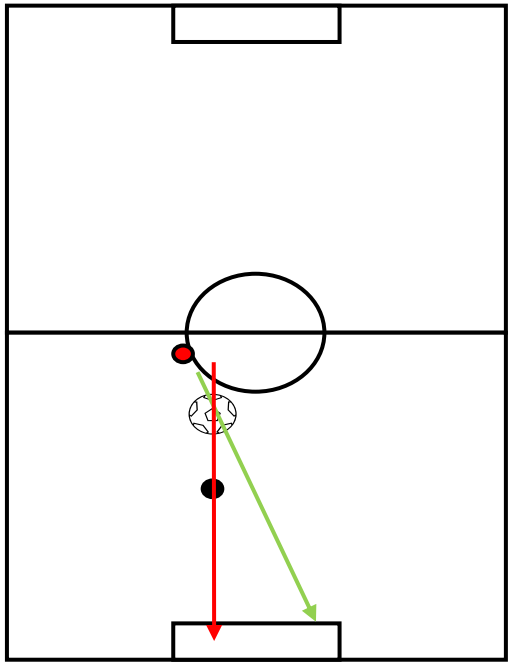
Fixed offsets fail to maintain accuracy at varied approach angles.

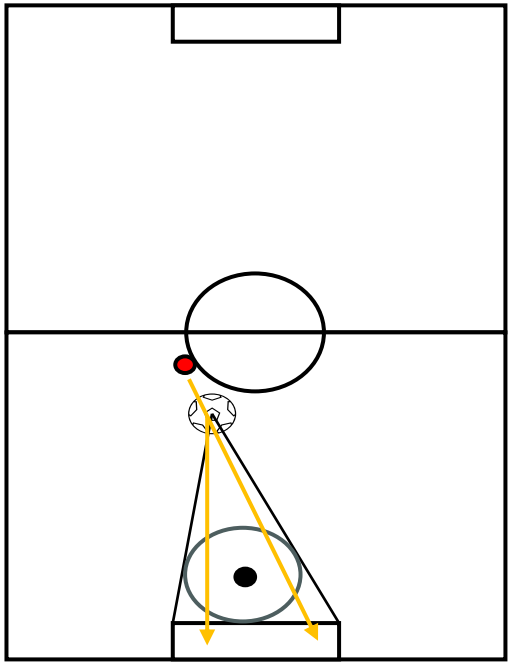
Supervised Learning Module: Automated data collection of optimal offsets across 360° approach angles.

```
rel_x,rel_y,angle,distance
-0.2390,-0.1018,-3.0931,0.9530
-0.2563,-0.1084,-3.0653,1.0856
-0.2507,-0.1136,-3.0329,1.0787
-0.2599,-0.1316,-3.0333,0.7714
-0.2664,-0.1455,-3.0096,1.1496
-0.2076,-0.2019,-3.0017,0.8614
-0.2556,-0.0468,-3.0091,1.0771
-0.2539,-0.1107,-2.9960,1.1040
-0.2879,-0.1164,-3.0055,0.8204
-0.2557,-0.1432,-2.9819,1.1308
-0.2525,-0.1304,-3.0004,0.7715
-0.2200,-0.1254,-2.9940,0.9438
-0.2807,-0.0824,-3.0088,0.7916
-0.1820,-0.2061,-3.0051,0.6711
-0.2571,-0.0943,-3.0024,0.7516
-0.2650,0.0446,2.7567,0.7598
```

The stability in our 700+ kicking trials is directly derived from this robust dataset and model.

Defensive Defender: Optimal Positioning & Coverage





Baseline: Specialized for straight shots; vulnerable to diagonal attacks.

Optimized: Positioning at the "Incenter" for balanced coverage of all angles.

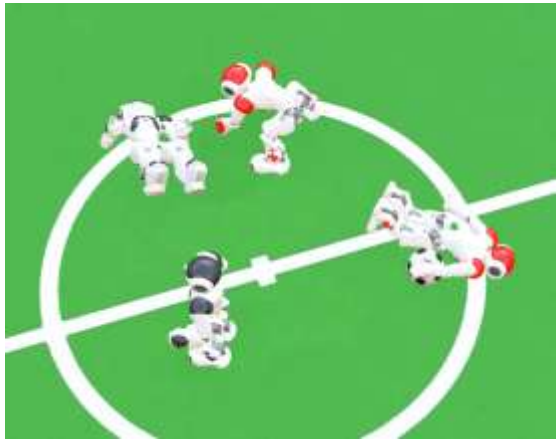
Test Scenario: 1v1 Duel. Success is defined as blocking the ball from crossing the defender's line.

Metric :Interception success rate		
	straight shots	diagonal attacks
Original	9/10	0/10
Improved	7/10	3/10

Strategic Choice: We avoided over-complex logic to prevent the defender from obstructing our Striker’s counter-attack path.

Attacking Defender: Synergy and Conflict Resolution

"Operational Conflict" (Identical logic leads to internal collisions).



"Role Differentiation" (Coordinated positioning prevents interference).

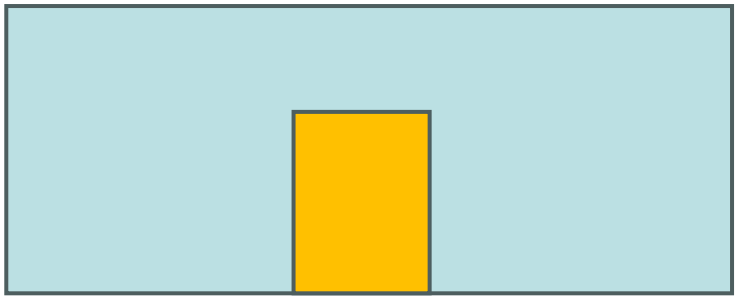
Benchmark: Team Offensive Efficiency (12min Match)		
	Total Goals (300s)	Time to First Goal
Symmetric Striker Logic (High Conflict)	1(12:00)	6:04(Red)
Black Team Single Striker Setup (Baseline)	2(11:30)	3:30s(Red)
Coordinated Multi-Role (Optimized)	0(12:00)	0:0

Scoring decreases when both sides deploy defenders; the four-robot conflict creates a 'traffic jam' that severely hinders offensive efficiency.

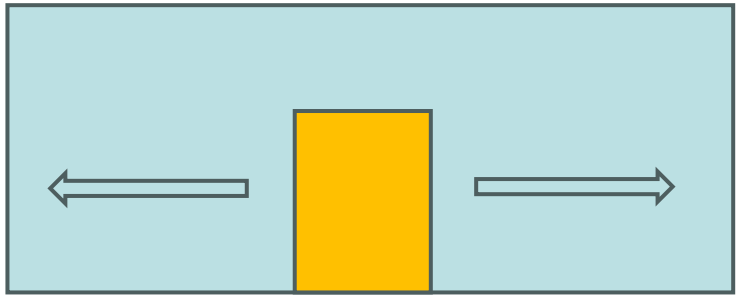
Numerical advantage (2 vs 1) typically provides an edge. Although an uncoordinated defender may obstruct its own team's offense, its defensive presence remains strategically valuable.

In the 'Coordinated' setup, the 0:0 result proves that the defender successfully neutralized all opponent attacks, demonstrating maximized defensive reliability at the cost of aggressive scoring.

Goalkeeper Evolution: From Static to Dynamic Defense

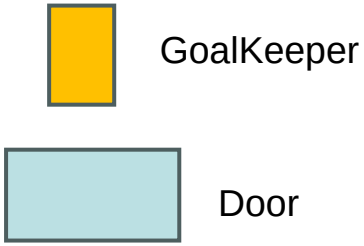


Phase 1: Static Centering.
Fixed position in the center. High risk for angled shots.



Phase 2: Lateral Tracking.
Synchronized Y-axis movement.
Successfully blocks most low-velocity shots.

Phase 3: Diving Save.
Implementation of lateral "leap" motions.
Maximizes coverage for high-speed corner shots.



Note: The diving motion is adapted from external open-source contributions. It is currently kept as a strategic reserve; since the current offensive threat has not yet reached the threshold requiring extreme saves, we prioritized system stability in the main project.
Link：https://github.com/sidharthsp/TDP_TEAM_11/tree/main